

Logic Design and Switching Theory

Lecture – 13

Subtitle: Sequential Circuit Design

by
Ms. Hera Noor

Previous

Shift Registers

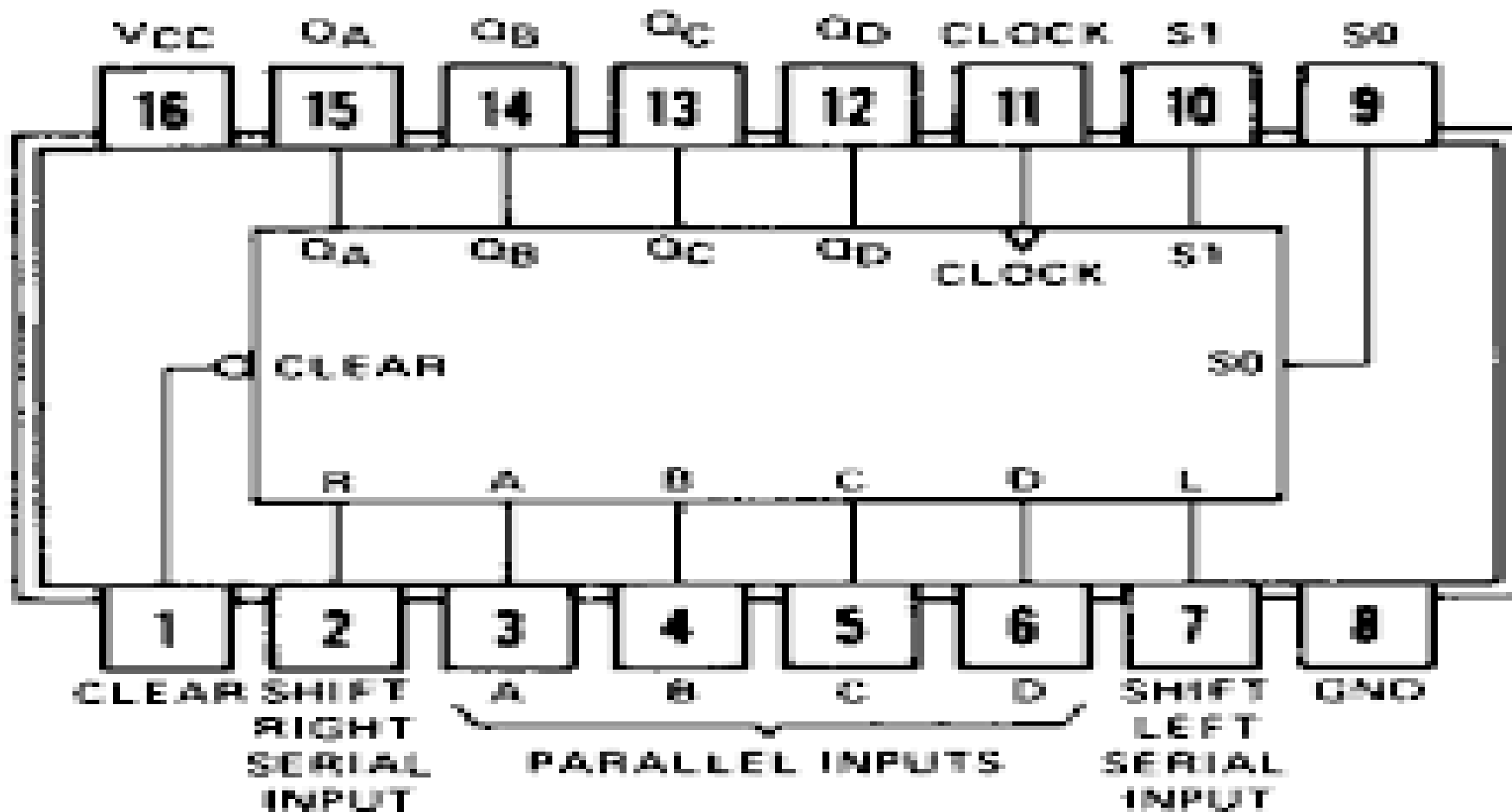
Today's Topic

Sequential Circuit Design

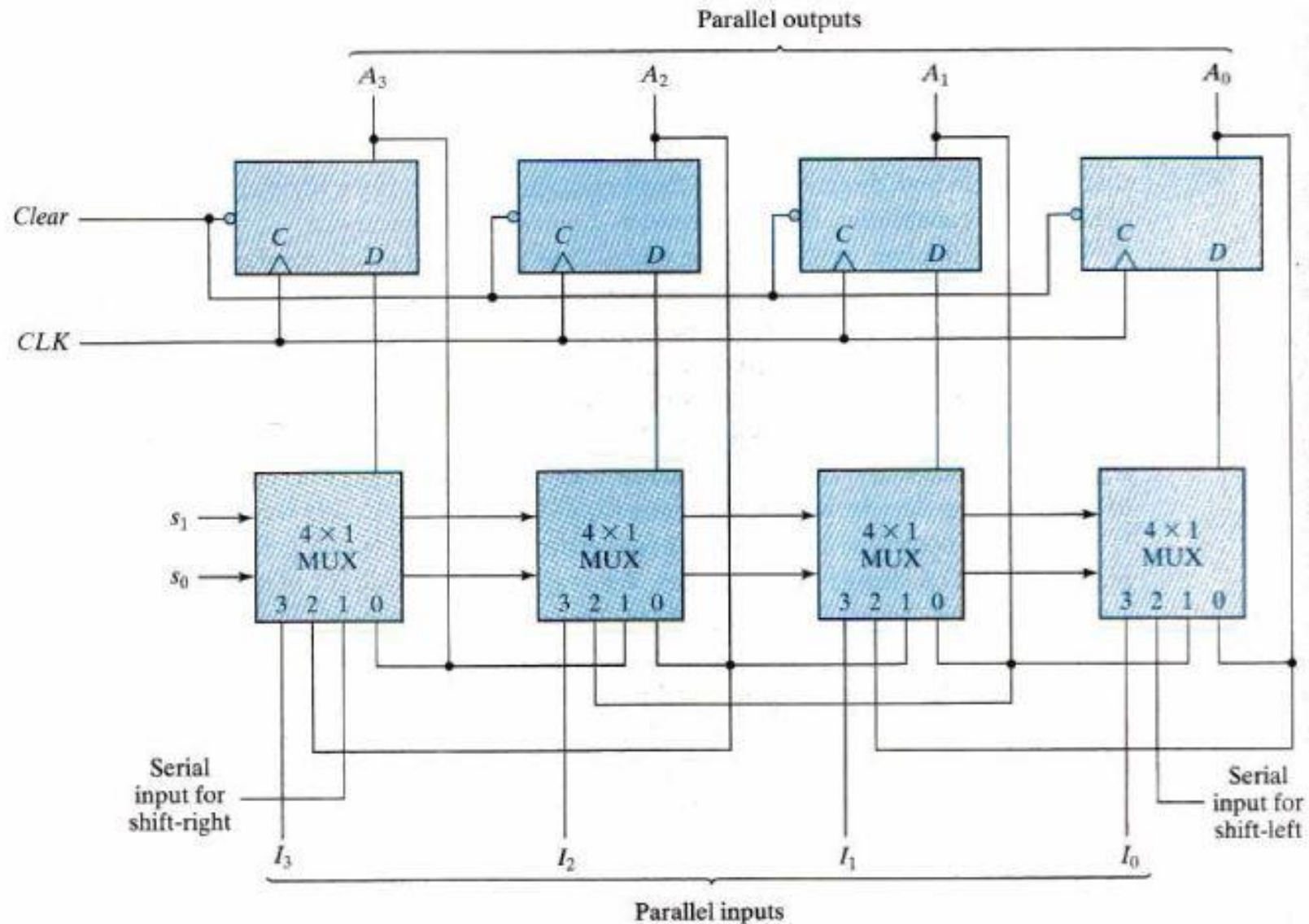
- Moore
- Mealy machine

Bidirectional Universal Shift Register

74194



Bidirectional Universal shift register



Bidirectional Universal Shift Register

CIR	S _i	S _o	CIR	Input	Output	
				A B C D	Q _A Q _B Q _C Q _D	
L	X	X	X	x x x x	L L L L	→ clear
H	X	X	L	x x x x	Q _A ... Q _D	→ retain previous state
H	H	H	↑	a b c d		→ parallel load ✓
H	L	H	↑	x x x x	H Q _A Q _B Q _C	} shift right
H	L	H	↑	x x x x	L Q _A Q _B Q _C	
H	H	L	↑	x x x x	Q _D ... Q _B H	} shift left
H	H	L	↑	x x x x	Q _B ... Q _D L	
H	L	L	↑	x x x x	Q _A - Q _D	→ Hold

Sequential circuit Design

Designing sequential circuits which store and process state over time, is central to digital system design.

A **state diagram** is a visual, graphical representation of a system's states and transitions, using circles and arrows to show how a sequential circuit or system moves from one state to another. A **state table** represents this same information tabularly, listing current states, inputs, next states, and outputs. Both are used in logic design to define how circuits change over time.

This involves finite state machines (FSMs), including two primary types: Moore and Mealy machines.

Sequential circuit Design

Step 1. State diagram

Step 2. State Table

Step 3. Reduction of states (in case of duplicate states).

Step 4. Assigning letters of symbols (if applicable).

Step 5. Determine the number of FFs.

Step 6. Derive the circuit excitation table.

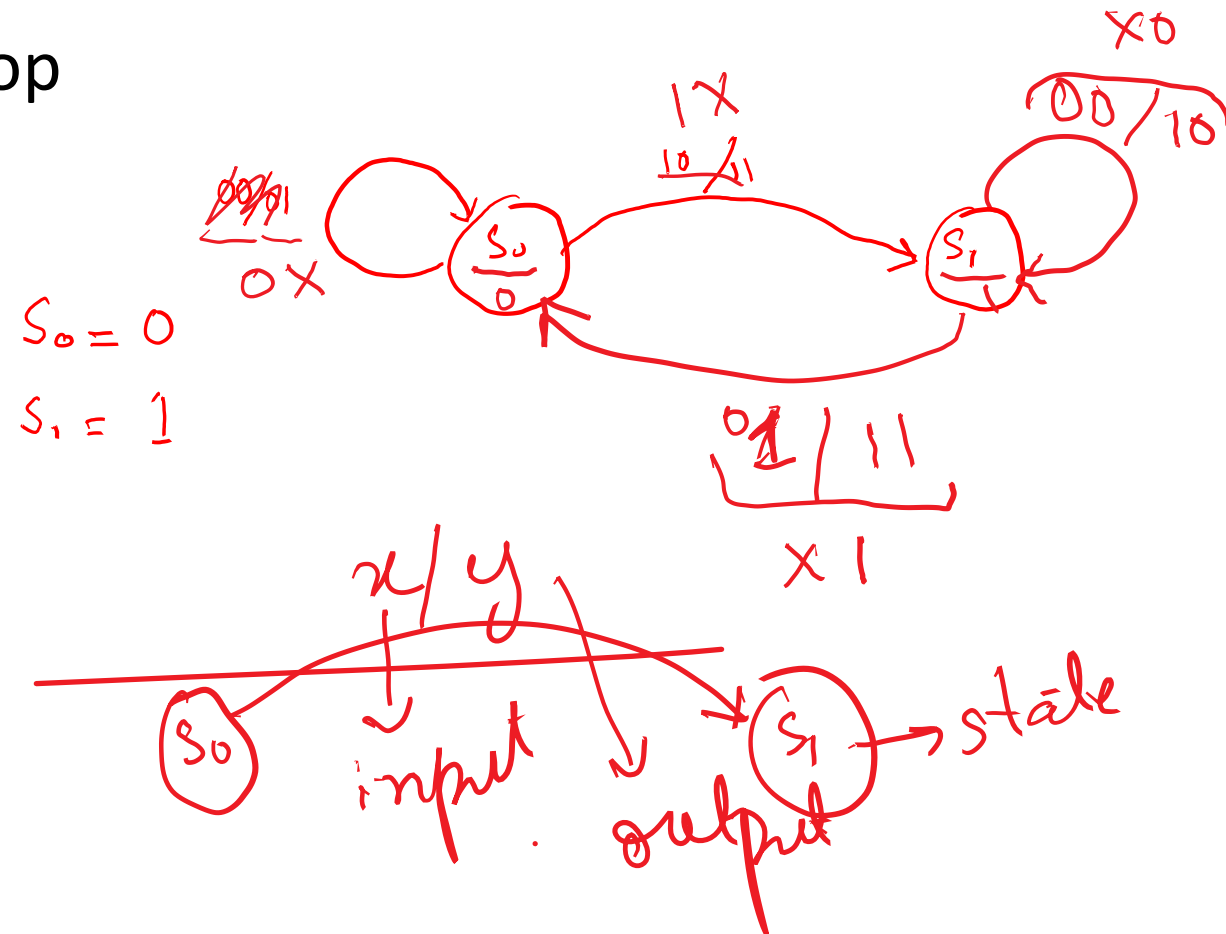
Step 7. Obtain equation of output and input for FFs using k-map.

Step 8. Design the circuit using FFs and gates according to question.

Sequential circuit Design

Develop the state diagram for JK flipflop

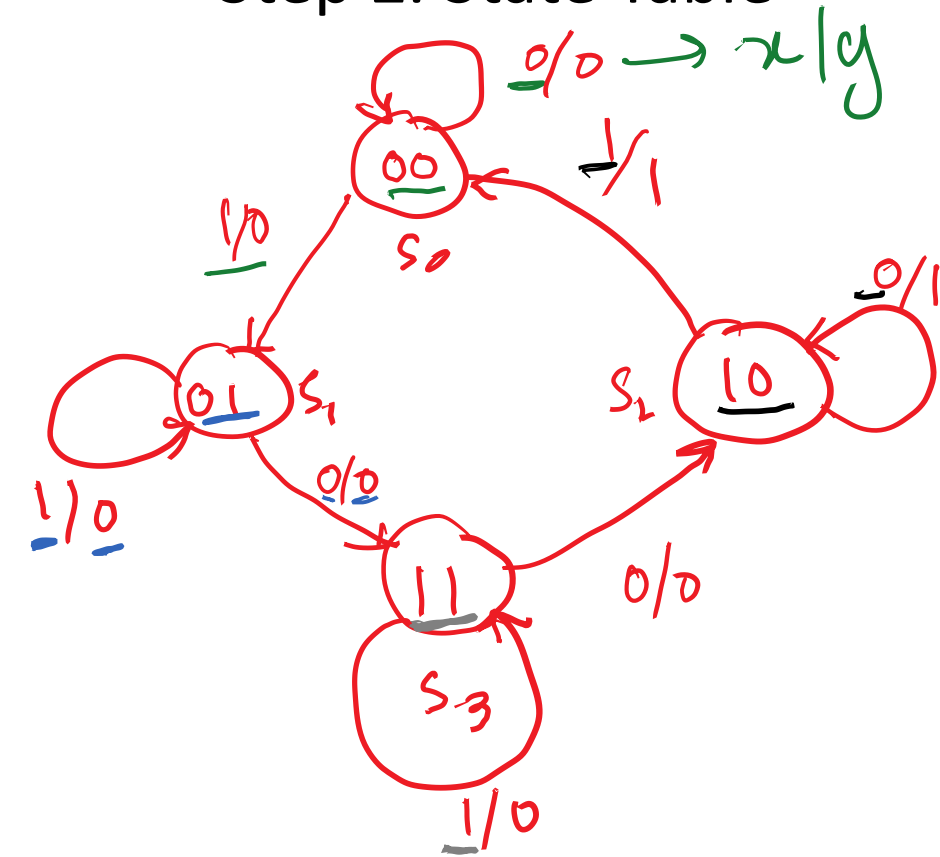
Q_t	J	k	Q_{t+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0



Example 2 - Sequential circuit Design

Step 1. State diagram

Step 2. State Table



P.S		Input (x, s)				O/P y	
Q _A	Q _B	x=0		x=1		x=0	x=1
		0	0	0	1	0	0
0	1	1	1	0	1	0	0
1	0	1	0	0	0	1	1
1	1	1	0	1	1	0	0

Step 5

4 → states

$2^2 \rightarrow n$

$n=2$

Sequential circuit Design

Step 5. Determine the number of FFs.

Step 6. Derive the circuit excitation table.

Q_A	Q_B	x	Q_A^+	Q_B^+	Input		y
					1-FF	2-FF	
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0
0	1	0	1	1	1	0	0
0	1	1	0	1	0	0	0
1	0	0	0	0	0	0	1
1	0	1	0	0	0	0	0
1	1	0	1	1	0	0	0
1	1	1	1	1	0	0	0

K-map

Q_m	Q_{m+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Sequential circuit Design

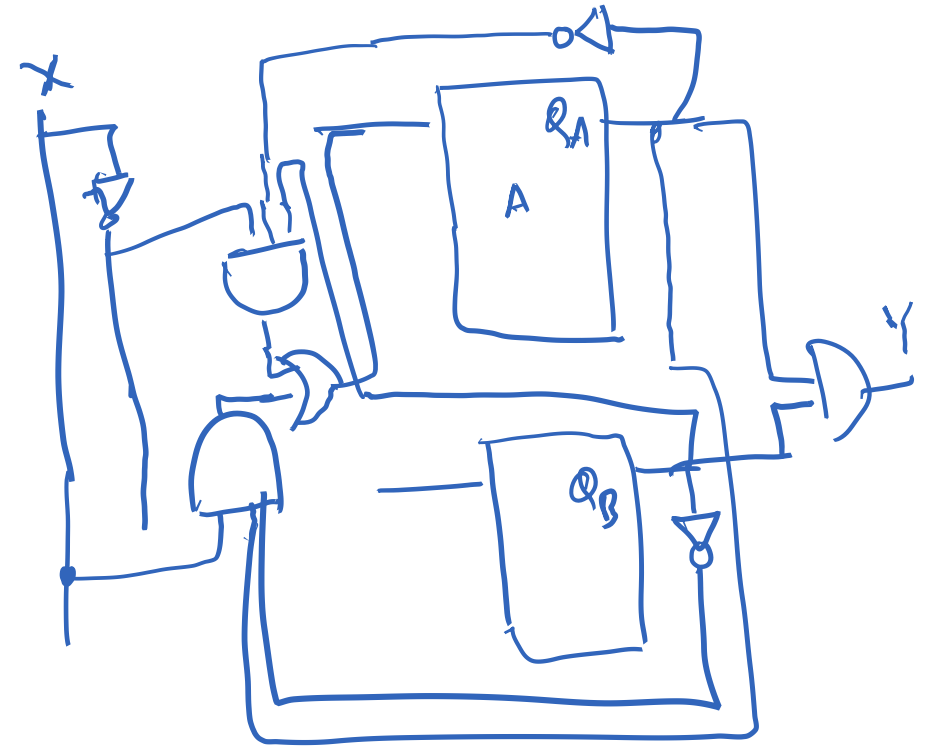
Step 7. Obtain equation of output and input for FFs using k-map.

Step 8. Design the circuit using FFs and gates according to question.

$$\begin{array}{ll} 1-FF & I/P = \overline{Q_A} Q_B \overline{x} + Q_A \overline{Q_B} x \\ 2-FF & I/P = \overline{Q_A} \overline{Q_B} x + Q_A Q_B \overline{x} \end{array}$$

output

$$y = Q_A \overline{Q_B}$$



FSM

FSMs model sequential logic using a finite number of states. Transitions between states depend on inputs, and FSMs produce outputs. FSMs are categorized into:

1. Moore Machine

- Output depends only on current state.
- Output is stable between clock edges.
- Simpler output logic, more predictable.

2. Mealy Machine

- Output depends on both current state and input.
- Output can change immediately when input changes.
- Can have fewer states than Moore (more compact design).

Moore and Mealy Machine

Mealy machine: created by G. H. Mealy, 1955

Moore machine: created by E.F. Moore, 1956

Purpose: to design a mathematical model for sequential circuits.

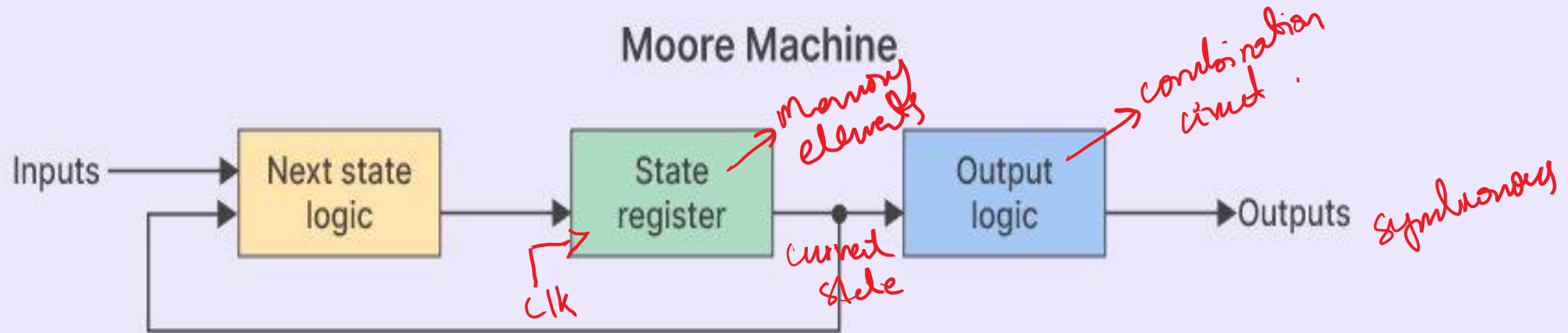
Definition: A Moore machine is a collection of five things

1. A finite set of states $q_0, q_1, q_2, \dots, q_n$ is the start state.
2. An alphabet of letters for forming the input string $\Sigma = \{a, b, c, \dots\}$
3. An alphabet of possible output characters $\Gamma = \{x, y, z, \dots\}$
4. A transition table that shows for each state and each input letter what state is reached next.
5. An output table that shows what character from Γ is printed by each state as it is entered

Moore Machine

- When each state of a finite state machine has an output symbol, the automata is known as a Moore machine. It generates an equivalent string for each input string.
- It includes all string transitions for the given alphabet. The current state and current input symbol determine the next state. After the entire input sequence has been applied, the output is evaluated.
- The output and state of the device change is synchronous with the clock edge. Moore machine generates the output associated with the initial state by reading no input symbol.
- The output of a Moore circuit is solely determined by the current state of the memory elements. If the length of the input string is n , the length of the string produced by this machine will be $(n+1)$.

Moore Machine



$M = (Q, \Sigma, \Delta, \delta, \Lambda, q_0)$ is a 6 tuple notation

Where Q is the set of finite states.

Σ is an input alphabet containing a finite number of input symbols

Δ is an output alphabet that contains a finite number of the output string

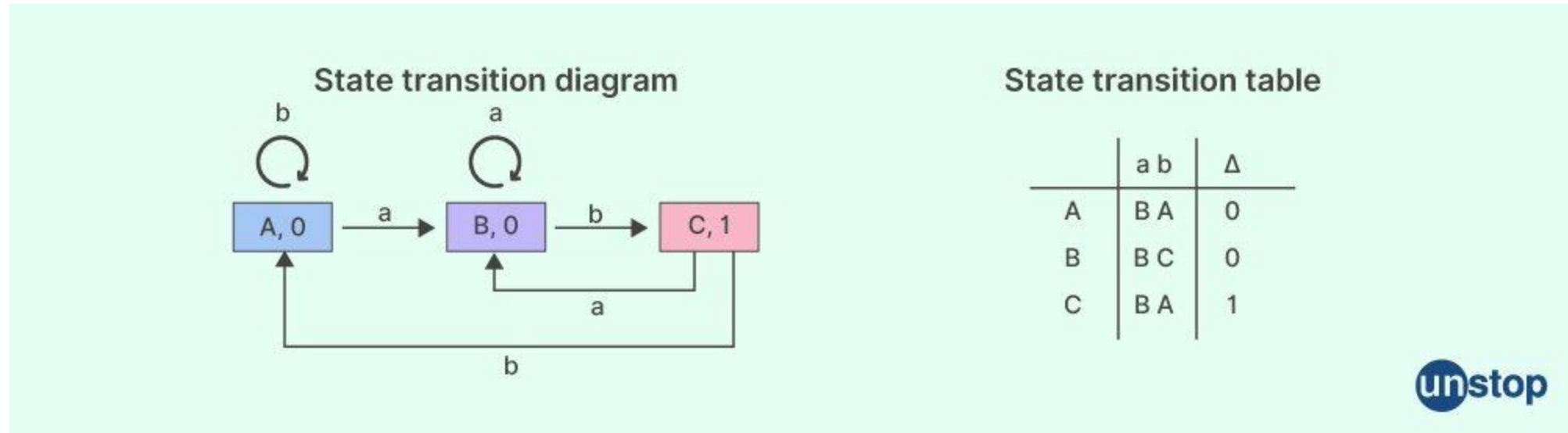
δ is a transition function of input $\delta: Q \times \Sigma \rightarrow Q$

Λ is an output function: $\Lambda: Q \rightarrow \Delta$ (output associated with state)

q_0 is an initial state, $q_0 \in Q$

Moore Machine

Construct a Moore machine that takes a set of all strings over $\{a, b\}$ as input and prints '1' as output for every occurrence of 'ab' as a substring.



$\Sigma(\text{input alphabet}) = (a, b)$

$\Delta(\text{output alphabet}) = (0, 1)$

$Q(\text{finite states}) = (A, B, C)$

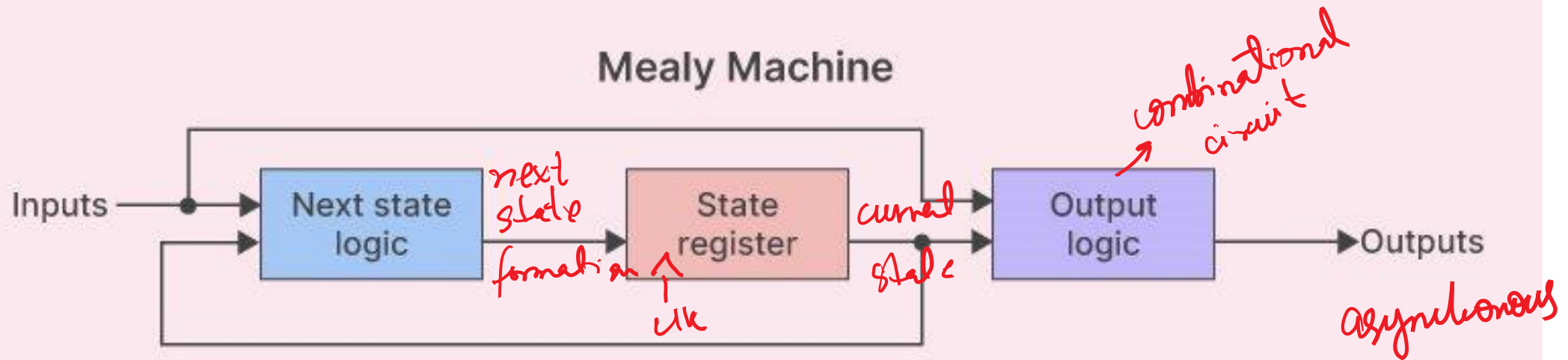
Input: abaababaa

Output: 0010010100

Mealy Machine

- A Mealy machine is a machine in the theory of computation in which the output symbol is linked to the transition (state and input) of a finite state machine. It creates an equivalent output string for each input letter/string.
- If the length of the input string is n , the length of the Moore machine's output string is also n . The value of the output function becomes a function of transitions and changes when the input logic is completed in the current state.
- On the current clock, the asynchronous output generation changes its state to synchronous. The Mealy circuit's output values are influenced by both the current state output of memory elements and external inputs.
- In this type of circuit, Mealy machines respond to inputs more quickly. They all seem to react in the same clock cycle. It takes care of all string transitions across a specified alphabet. A Mealy machine is not a counter.

Mealy Machine



Mealy Machine Characteristics

$M = (Q, \Sigma, \Delta, \delta, \Lambda, q_0)$ is a 6 tuple notation

Where Q is the set of finite states.

Σ is an input alphabet containing a finite number of input symbols.

Δ is an output alphabet which contains a finite number of output string

δ is a transition function $\delta: Q \times \Sigma \rightarrow Q$

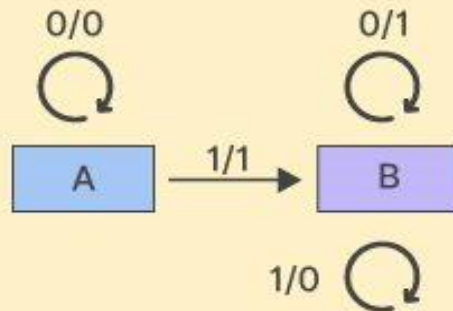
Λ is an output function: $\Lambda: Q \times \Sigma \rightarrow \Delta$ (output associated with state and input)

q_0 is an initial state, $q_0 \in Q$.

Mealy Machine

Construct a mealy machine that generates 2's complement of a binary number.

State transition diagram



State transition table

	0		1	
	State	Output	State	Output
A	A	0	B	1
B	B	1	B	0

$\Sigma(\text{input alphabet}) = (0, 1)$

$\Delta(\text{output alphabet}) = (0, 1)$

$Q(\text{finite states}) = (A, B)$

Input = 1100

Output = 0100

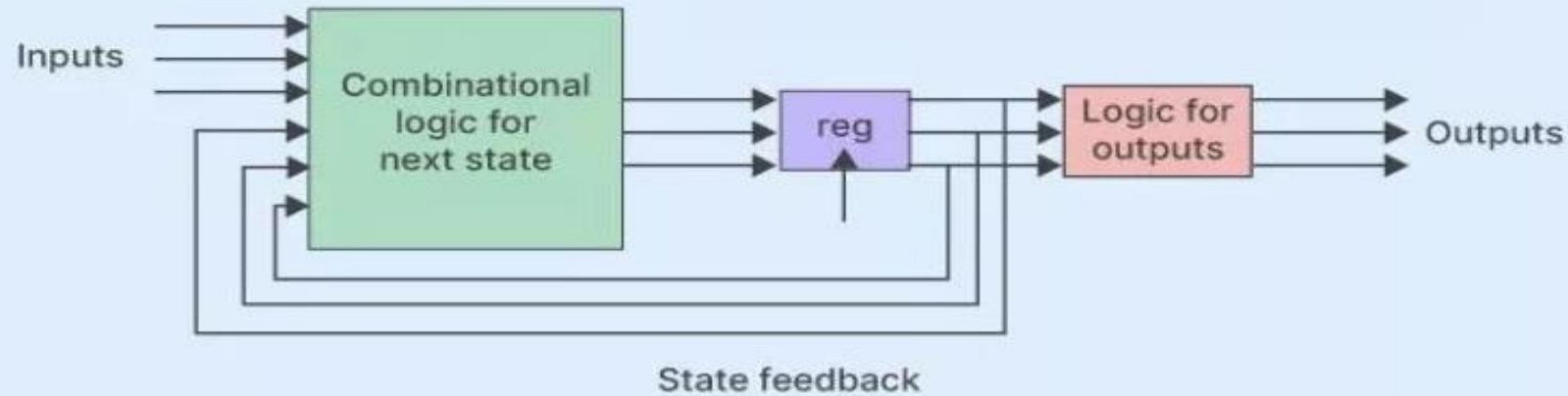
Read I/p = 0011

Produced O/p = 0100

Mealy Machine	Moore Machine
1. The output of the circuit may be affected by changes in the information.	1. The output of the circuit is unaffected by changes in the input.
2. It requires fewer number of states for implementing the same function.	2. For implementing the same function, it requires more number of states
3. The output is a function of the current state as well as the input.	3. The output is a function of the current state only.
4. The counter cannot be referred to as a Mealy Machine.	4. The counter is referred to as a Moore Machine.
5. The design of the Mealy model is complex.	5. The design of the Moore model is easy.
6. Mealy machines react to changes more quickly. They all seem to react in the same way.	6. Decoding the output necessitates more logic, resulting in longer circuit delays. They usually react after one clock cycle
7. If the external inputs vary, the output can alter between clock edges.	7. Only when the active clock edge will the output alter.
8. The output is set on the transition.	8. The output is set to state.
9. It's easier to design with less hardware.	9. To design, more hardware is necessary

Mealy vs Moore Machine

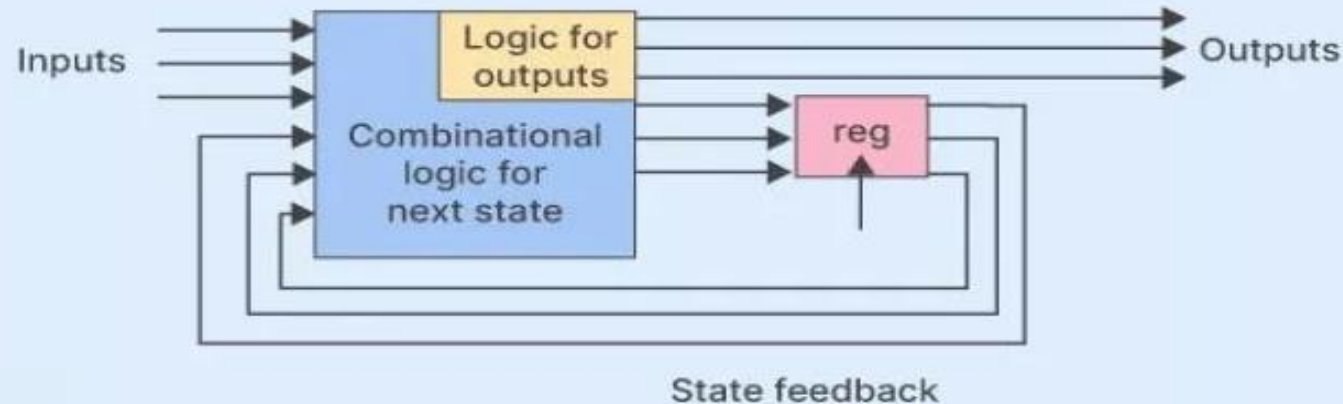
What's the difference?



Moore machine

Outputs are a function of current state

Outputs change synchronously with state changes



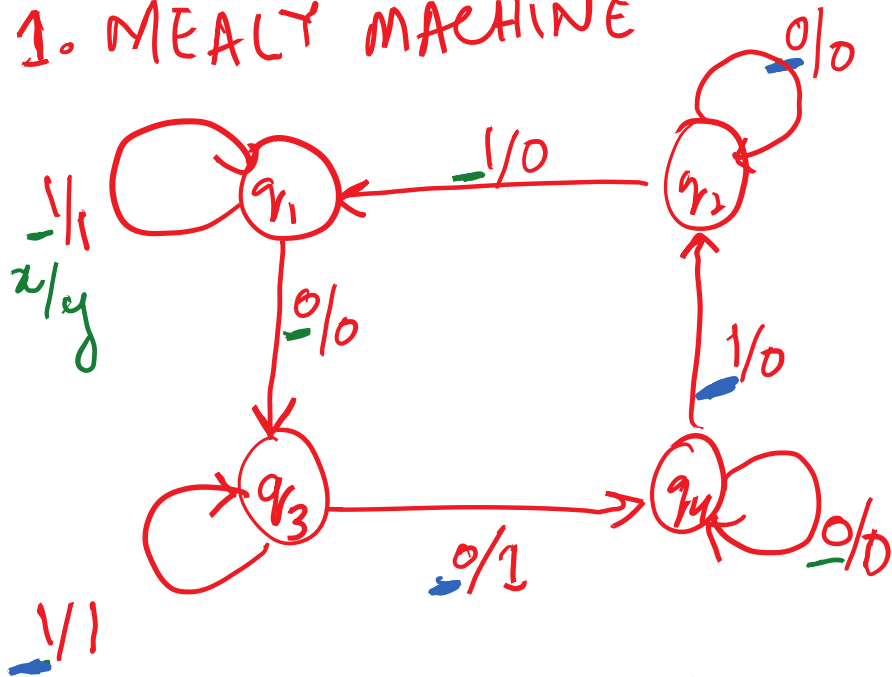
Mealy machine

Outputs depend on state and on inputs

Input changes can cause immediate output changes (asynchronous)

Conversion of State Diagram and Table with Input Sequence

1. MEALY MACHINE



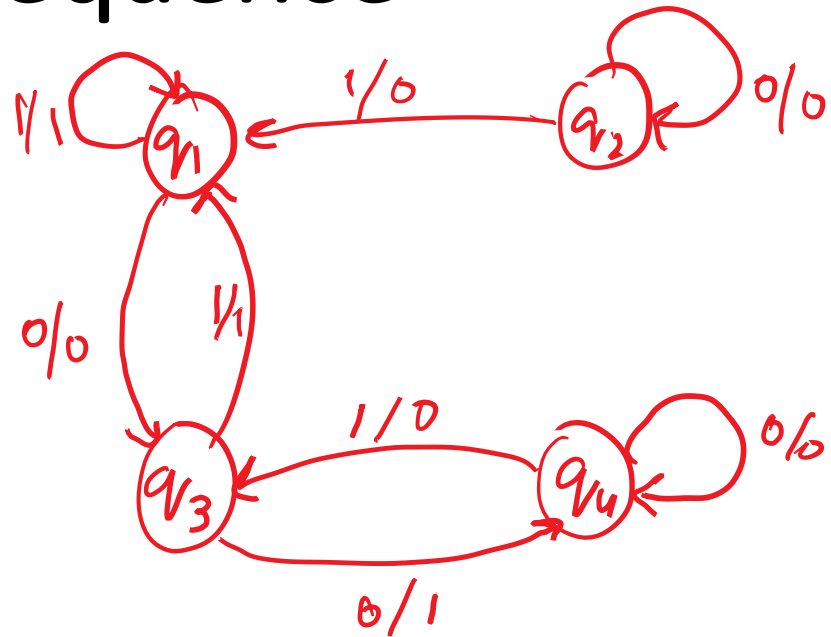
IS : 1010 to q_3
 OS : 1100 Ans

State table

$x=0$ $x=1$ $\rightarrow$ i/p n.s/o

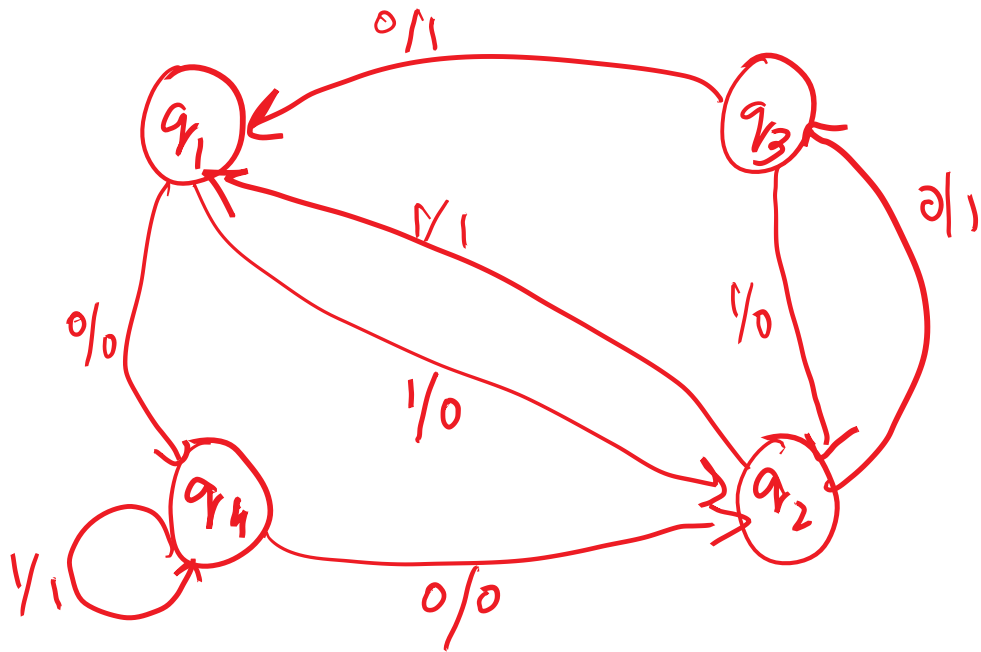
q_1	$q_3, 0$	$q_1, 1$
q_2	$q_2, 0$	$q_1, 0$
q_3	$q_4, 1$	$q_3, 1$
q_4	$q_4, 0$	$q_2, 0$

Conversion of State Diagram and Table with Input Sequence



IS : 0101 to q_1

Conversion of State Diagram and Table with Input Sequence



IS : 0110

OS : ?

Conversion of State Diagram and Table with Input Sequence

Conversion of State Diagram and Table with Input Sequence

Conversion of State Diagram and Table with Input Sequence

Thank you!